

Talking Nets: An Oral History of Neural Networks

edited by James A. Anderson and Edward Rosenfeld, MIT Press, 1998.

£31.95/\$39.95 (xi + 434 pages) ISBN 0 262 01167 0

What is it about neural networks that makes people so excitable?

James Anderson, a noted neural-network researcher, and Edward Rosenfeld, a journalist, had the splendid idea to generate an oral history of neural-net research by talking to most of the pioneering figures in the field. The result, *Talking Nets*, is an extraordinary document that is a page-turner for cognitive scientists.

Physics is a science. Entymology is a science. 'Neural networks' is a toolbox – a collection of tools for the simulation and analysis of complex systems. These tools were developed by people working in diverse fields ranging from control theory and signal processing to neurobiology and psychology. They are being vigorously applied to an equally broad range of problems ranging from understanding basic brain mechanisms to forecasting fluctuations in the bond market. This book recapitulates much of the convoluted history of neural-network research through the testimony of people who figured centrally in its development. It is a remarkable story, replete with drama, tragedy, hubris, irony, humor, bitter intellectual warfare, and a couple of corpses. It is told in the words of a brilliant and eccentric collection of people, including several certifiable geniuses and a couple who are merely certifiable, as Groucho would have said. (The interviewees comprise Bernard Widrow, Carver Mead, Stephen Grossberg, Michael Arbib, James Anderson, David Rumelhart, Geoff Hinton, Terry Sejnowski and nine others.) The accounts aren't all consistent with each other and can't all be equally true. Yet in reading them, one gets a vivid sense of where people came from, what they thought they were doing, and how their ideas developed. It's also a lot of fun.

At one level, the history of neural-network research can be read as a long-running soap opera with a cast of characters that is a little strange even for a group of academics. The approach originated with some papers by Turing (on biological dynamic systems), Wiener's 'cybernetics', and the crucial articles by McCulloch and Pitts treating neurons as computational devices, which in turn greatly influenced von Neumann, who was about to invent the modern computer. McCulloch was a neurophysiologist at the University of Chicago. Pitts was a prodigy and misfit who hung around the university but wasn't actually enrolled. The story of how Pitts met McCulloch via Carnap

and Bertrand Russell is worth the price of the book (for which you get several versions). The classic McCulloch and Pitts papers were written when Pitts was still in his teens and living with McCulloch and his wife. Pitts subsequently gravitated to a second father-figure, Norbert Wiener. Wiener and McCulloch had a falling out because Mrs Wiener didn't like McCulloch; Pitts was caught in the middle, had a nervous breakdown, and drank himself to an early grave. And we haven't even gotten to Minsky and Papert yet.

The tale of how Minsky and Papert slew the neural-net dragon in the 1960s is the most important legend in neural-net culture. The testimony in *Talking Nets* suggests a more complicated story, however. It is true that Minsky and Papert come off badly in this telling of the tale (of course, it isn't their side that is being told). The MIT people seem to have been bothered by publicity over neural-network research being conducted elsewhere (Widrow: 'You know, we had a lot of controversy in the early days. It was due to publicity that [Frank] Rosenblatt had in the news media and publicity that I had...I found that this kind of publicity infuriates colleagues...'). One has to imagine what transpired the day that Rosenblatt, the psychologist from Cornell, came down to MIT to tell an audience that included Minsky, Shannon and McCulloch the news about perceptrons. (Cowan: 'It was a terrible lecture. McCulloch didn't say anything. Shannon said, 'It's worth looking at'... But by and large it was clear that the perceptron wasn't doing the things that Frank claimed it could do.') Rosenblatt had the even greater misfortune to die soon thereafter in a boating accident.

The ironies here are staggering. Minsky's reaction to the publicity over perceptrons has to be considered in light of his subsequent willingness to serve the mass media as a kind of all-purpose, techno-egghead commentator on the issues of the day (e.g. 'The future of money', *Discover*, 1998; 'Is the body obsolete?', *Whole Earth Review*, 1989). And if the perceptron was not able to do all the things that were being claimed, how would that compare to the years of relentless hype about artificial intelligence?

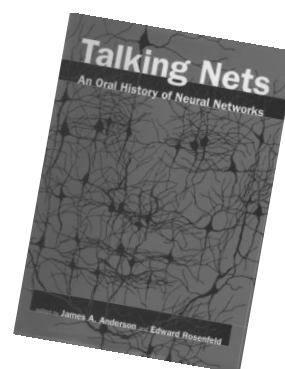
On the other hand, did it matter? Interestingly, opinions vary. Robert Hecht-Nielsen thinks that Minsky and Papert's book¹ *Perceptrons* mattered because it created 'a new conventional wisdom that some MIT professors have

proven mathematically, that neural networks cannot ever do anything interesting.' Widrow, in contrast, says, 'I couldn't understand what the point of [*Perceptrons*] was, why the hell they did it. I figured that they must have gotten inspired to write that book really early on to squelch the field, to do what they could to stick pins in the balloon. But by the time the book came out, the field was already gone. There was just about nobody doing it.'

So what really happened? It seems clear that people such as Widrow had become discouraged because they knew that multilayer nets were needed but didn't know how to train them. His testimony on this point is poignant:

'We would have given our eye teeth to come up with something like Backprop...Backprop to me is almost miraculous. The first exposure I had to Backprop was around 1985 at a meeting at Snowbird, Utah... Someone gave a paper in the first morning session and during the question period...someone got up and said, "You know, something like that was done by Widrow back in the early '60s." They began to have this big discussion about what Widrow did and Widrow didn't do, and I'm just sitting there, listening to all this stuff...I was like a dead man. I was a man who'd died, who was sitting up on a cloud somewhere, looking down on the Earth, watching what happened after he died.' (p. 61)

People also had to wait for computer power to catch up with the ideas, a point that Arbib emphasizes. Widrow and Rosenblatt were experimenting with physical networks created out of electric motors and potentiometers (some good to three decimal places!). Computers were primitive, limiting the use of simulation as a tool; luckily things were to move ahead rapidly on that front. In the meantime, some interesting work was done and the people in this book managed to survive quite well, ending up with positions at places like Berkeley, Stanford, and CalTech.



Other people had a rockier time, however. Consider the harrowing tale of Paul Werbos, who solved the problem that stymied Widrow by inventing Backprop (he says he was led to the idea by an interest in 'translating Freud into mathematics'): Werbos experienced the classic graduate student's nightmare, a thesis committee that didn't appreciate the significance of what he had done. He was informed that he wouldn't pass unless he found a patron to attach his imprimatur to the research. As Werbos tells it, he went to Grossberg (then an assistant professor at MIT) who told him, 'This stuff you've done, it's already been done before. Or else it's wrong. I'm not sure which of the two, but I know it's one of the two.' He went to Minsky, who said he couldn't get involved. He went to Jerome Lettvin (co-author of 'What the frog's eye tells the frog's brain'), who said he had the wrong idea about how the brain worked. So he couldn't find a patron. He finally got a degree by doing simulations of existing data. He eventually drifted to the National Science Foundation where he became a program director, which is one way to see to it that other people get more support for interesting research than he did (surely the most Freudian aspect of the story). His work was so buried in the literature that the algorithm was subsequently rediscovered several times, most notably by Rumelhart and colleagues.

Some of the other people who left the field also managed to lead constructive lives. Hoff, of the Widrow-Hoff (least-mean-squared) learning algorithm, for example, went to Intel where he invented the microprocessor. Still, for a time the Minsky and Papert book does seem to have had enormous impact on what work was considered worth doing. The testimony in *Talking Nets* suggests that Minsky and Papert

were highly effective speakers who spent considerable time spreading their message. Their personal brilliance and charisma, combined with the aggressive way the results were presented, were apparently sufficient to persuade people to believe what had not been proved (and turned out to be wrong) – that similar limitations would be observed with more elaborate architectures.

The other issue that permeates the book is the 'credit assignment problem', that is, who discovered what (this is a small ironic joke because of the original sense of the term, which concerns the difficulty of analysing how a network has solved a problem). Several people describe obscure papers in which they claim to have invented one or another kind of net, learning algorithm, or representational scheme, now primarily known through someone else's work. Several people are more concerned that other people not be given undue credit; the main focus of this sentiment is John Hopfield, because the architecture now universally known as the Hopfield net is said to have been invented by others. Everyone has a different story about where the central ideas came from; Lettvin insists that it all goes back to Leibniz.

I don't know if these conflicting claims could be adjudicated without convening the World Court. Reading this book, however, makes it clear that few of these individuals are prepared to take the longer view that some of the ambiguity about intellectual priority derives from non-pathological factors to do with the conditions under which the neural-network approach evolved. People were working in vastly different fields, including engineering, psychology, physics and biology. Who would have been routinely scanning all of the literature that turned out to be relevant, in that pre-web world? The

conditions made it possible for someone like Widrow, who presented the LMS algorithm in the 'Institute of Radio Engineers, Western Electronic Show and Convention, Convention Record, Part 4', to be unaware of Werbos' unpublished Harvard thesis titled 'Beyond regression: new tools for prediction and analysis in the behavioral sciences.' Under these conditions convergent evolution can also occur: people who were trying to solve different kinds of problems (building an adaptive analogue filter, a model of a neuron) ended up discovering the same tools. It is also true that putting existing elements together in novel ways or explaining something well and connecting it to things people already understand are important original contributions.

Some historian of science will eventually recognize what a fascinating story this is and perhaps be able to track down where the really essential insights originated (have the system adjust itself!; let the units participate in different representations!) and how the different strands were brought together. In the meantime we have this wonderful book. The editors have done an outstanding job of editing the interviews, which read well while preserving the interviewees' own voices. Get hold of this book and spend a couple of days with a remarkable group of people and the story of their prodigious intellectual achievements.

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Reference

- 1 Minsky, M. and Papert, S. (1969) *Perceptrons*, MIT Press

Books Received

Review copies of the following books have been received. Books that have been reviewed in Trends in Cognitive Sciences are not included.

The appearance of a book in the list does not preclude the possibility of it being reviewed in the future.

E. Basar (ed.) *Brain Function and Oscillations II: Integrative Brain Function. Neurophysiology and Cognitive Processes* Springer, 1999. DM 148.00/£57.00/\$89.95 (xxxix + 476 pages) ISBN 3 540 64345 1

T.A. Brown (ed.) *Molecular Biology Labfax: I. Recombinant DNA (second edition)* Academic Press, 1998. \$59.95 (xxviii + 377 pages) ISBN 0 12 136055 5

R. Cotterill *Enchanted Looms: Conscious Networks in Brains and Computers*

Cambridge University Press, 1998. £17.95/\$24.95 (xiv + 508 pages) ISBN 0 521 62435 5

C. Koch *Biophysics of Computation: Information Processing in Single Neurons* Oxford University Press, 1999. £59.95 (xxiii + 562 pages) ISBN 0 19 510491 9

E.M. Macphail *The Evolution of Consciousness* Oxford University Press, 1998. PB £15.99, HB £39.99 (xiv + 256 pages) ISBN PB 0 19 850324 5, HB 0 19 850325 3

V.B. Mountcastle *Perceptual Neuroscience: The Cerebral Cortex* Harvard University Press, 1998. £37.50 (xvii + 486 pages) ISBN 0 674 66188 5

J. Panksepp *Affective Neuroscience: The Foundations of Human and Animal Emotions* Oxford University Press, 1998. £49.50 (xii + 466 pages) ISBN 0 19 509673 8

R.W. Parks, D.S. Levine and D.L. Long (eds) *Fundamentals of Neural Network Modeling: Neuropsychology and Cognitive Neuroscience* The MIT Press, 1999. £41.95 (xii + 428 pages) ISBN 0 262 16175 3

M. Zigmond, F. Bloom, S. Landis, J. Roberts and L. Squire (eds) *Fundamental Neuroscience* Academic Press, 1999. (xvi + 1600 pages) ISBN 0 12 780870 1